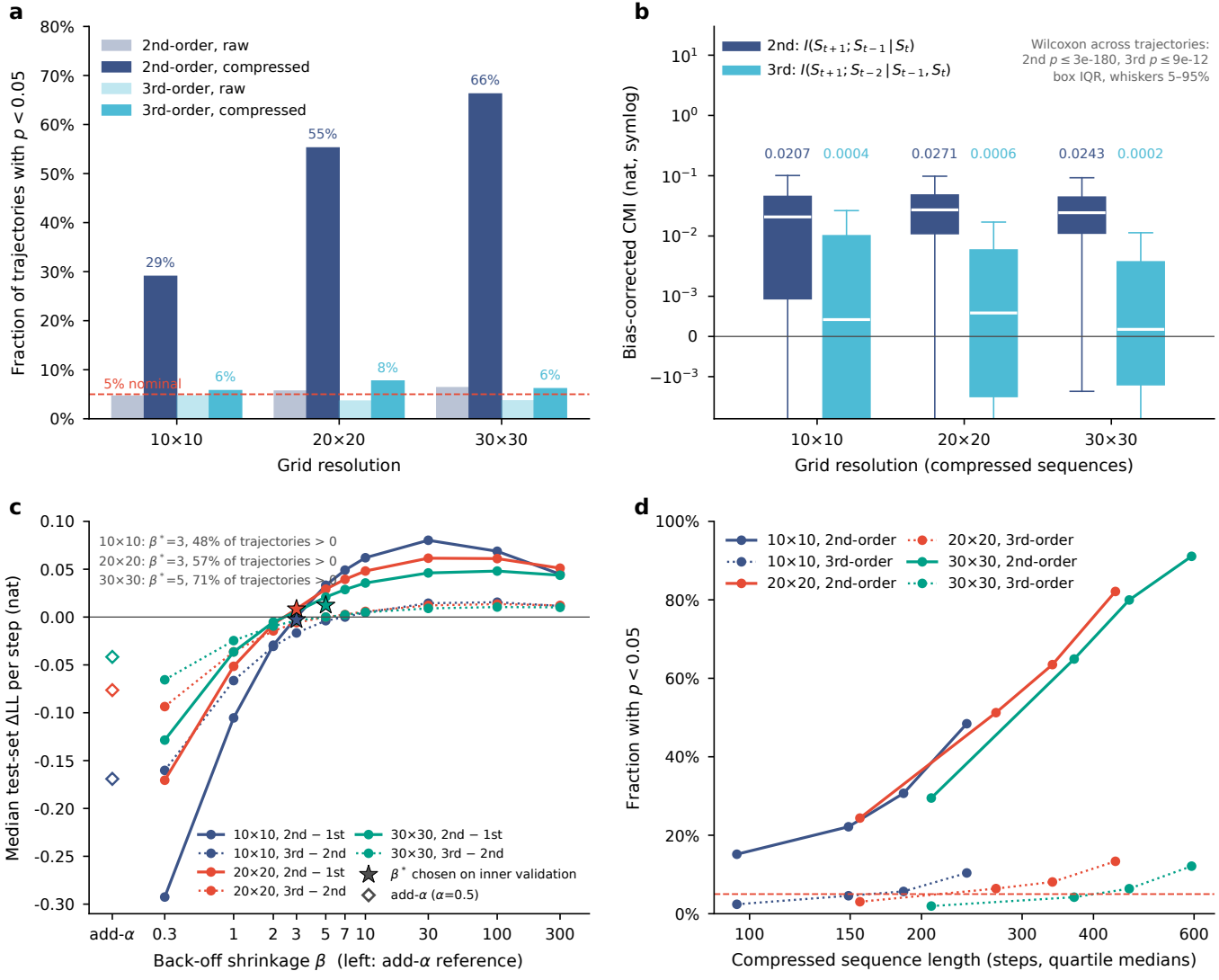
a
Fraction of trajectories with $p < 0.05$
2nd-order, raw
2nd-order, compressed
3rd-order, raw
3rd-order, compressed
5% nominal
29%
6%
55%
8%
66%
6%
10×10
20×20
30×30
Grid resolution
b
2nd: $I(S_{t+1}; S_{t-1} \mid S_t)$
3rd: $I(S_{t+1}; S_{t-2} \mid S_{t-1}, S_t)$
Wilcoxon across trajectories:
2nd $p \le$ 3e-180, 3rd $p \le$ 9e-12
box IQR, whiskers 5–95%
Bias-corrected CMI (nat, symlog)
0.0207
0.0004
0.0271
0.0006
0.0243
0.0002
Grid resolution (compressed sequences)
c
Median test-set ΔLL per step (nat)
10×10: β^*=3, 48% of trajectories > 0
20×20: β^*=3, 57% of trajectories > 0
30×30: β^*=5, 71% of trajectories > 0
10×10, 2nd − 1st
10×10, 3rd − 2nd
20×20, 2nd − 1st
20×20, 3rd − 2nd
30×30, 2nd − 1st
30×30, 3rd − 2nd
β^* chosen on inner validation
add-α (α=0.5)
add-α
Back-off shrinkage β (left: add-α reference)
d
Fraction with $p < 0.05$
10×10, 2nd-order
10×10, 3rd-order
20×20, 2nd-order
20×20, 3rd-order
30×30, 2nd-order
30×30, 3rd-order
Compressed sequence length (steps, quartile medians)